

Total No. of Questions : 5]

SEAT No. :

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[6432]-15

First Year M.C.A. (Management)
MT - 21 : OPTIMIZATION TECHNIQUES
(Revised 2020 Pattern) (Semester-II)

Time : 2½ Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) All questions are compulsory.*
- 2) Use of statistical table and non-programable calculator is allowed.*
- 3) Figures to the right indicate full marks.*

Q1) Use the simplex method to solve the following LP problem. **[10]**

Maximize $Z = 3x_1 + 5x_2 + 4x_3$ Subject to the constraints

$$2x_1 + 3x_2 \leq 8,$$

$$2x_2 + 5x_3 \leq 10,$$

$$3x_1 + 2x_2 + 4x_3 \leq 15 \text{ and } x_1, x_2, x_3 \geq 0$$

OR

Solve the following L.P.P. by two-phase method **[10]**

Maximize $Z = 4x + 3y$ Subject to the constraints

$$2x + 3y \leq 6$$

$$3x + 1y \geq 3 \text{ Both } x \text{ and } y \text{ all } \geq 0$$

- Q2)** a) Two manufacturers A and B are competing with each other in a restricted market. Over the year, A's customers have exhibited a high degree of loyalty as measured by the fact that customers are using A's product 80 per cent of the time. Also former customers purchasing the product from B have switched back to A's product 60 per cent of the time. **[7]**
- i) Construct and interpret the state transition matrix in terms of (1) retention and loss, and (2) retention and gain
 - ii) Calculate the probability of a customer purchasing A's product at the end of the second period.
- b) Explain the any one quantitative method that is useful for decision-making under uncertainty with example. **[3]**

OR

P.T.O.

- a) A bakery keeps stock of a popular brand of cake. Previous experience shows the daily demand pattern for the item with associated probabilities, as given below: [7]

Daily demand (number) :	0	10	20	30	40	50
Probability :	0.01	0.20	0.15	0.50	0.12	0.02

Use the following sequence of random numbers to simulate the demand for next 10 days.

Random numbers : 25, 39, 65, 76, 12, 05, 73, 89, 19, 49.

Also estimate the daily average demand for the cakes on the basis of the simulated data.

- b) Explain the following essential components of decision model : [3]

- Decision alternatives
- States of nature
- Payoff

- Q3)** a) A readymade garments manufacturer has to process 7 items through two stages of production, viz., cutting and sewing. The time taken for each of these items at the different stages are given below in appropriate units:[7]

Item :	1	2	3	4	5	6	7
Process time Cutting :	5	7	3	4	6	7	12
Sewing :	2	6	7	5	9	5	8

Find an order in which these items are to be processed through these stages so as to minimize the total processing time.

- b) ABC Corporation wants to launch one of its mega campaigns to promote a special product. The promotion budgets not yet finalized, but they know that some Rs. 55,00,000 is available for advertising and promotion. Management wants to know how much they should spend for television spots, which is the most appropriate medium for their product. They have created five 'T.V. campaign strategies' with their projected outcome in terms of increase in sales. Find which one they have to select to yield maximum utility. The data required is given below. [3]

Strategy	Cost in lakhs in Rs.	Increased in sales in lakhs of Rs.
A	1.80	1.78
B	2.00	2.02
C	2.25	2.42
D	2.75	2.68
E	3.20	3.24

OR

- a) A machine operator has to perform three operations, namely plane turning, step turning and taper turning on a number of different jobs. The time required to perform these operations in minutes for each operating for each job is given in the matrix given below. Find the optimal sequence, which minimizes the time required. [7]

Job	Time for plane turning in minutes	Time for step turning in minutes	Time for taper turning in minutes
1	3	8	13
2	12	6	14
3	5	4	9
4	2	6	12
5	9	3	8
6	11	1	13

- b) In a game of head and tail of coins the player A will get Rs. 4/- when a coin is tossed and head appears; and will lose Rs. 5/- each time when tail appears. Find the optimal strategy of the player. [3]

Q4) A small project is composed of 7 activities whose time estimates are listed below. Activities are being identified by their beginning (i) and ending (j) node numbers. [10]

Activities		Time in weeks		
i	j	t_0	t_l	t_p
1	2	1	1	7
1	3	1	4	7
1	4	2	2	8
2	5	1	1	1
3	5	2	5	14
4	6	2	5	8
5	6	3	6	15

- Draw the network
- Calculate the expected variances for each
- Find the expected project completed time
- Calculate the probability that the project will be completed at least 3 weeks than expected
- If the project due date is 18 weeks, what is the probability of not meeting the due date

OR

The utility data for a network are given below. Determine the total, free, independent and interfering floats and identify the critical path. [10]

Activity :	0-1	1-2	1-3	2-4	2-5	3-4	3-6	4-7	5-7	6-7
Duration :	2	8	10	6	3	3	7	5	2	8

- Q5) a) Solve the given game by method of matrices: [7]

		B		
		I	II	III
A	I	1	-1	-1
	II	-1	-1	3
	III	-1	2	-1

- b) An investor is given the following investment alternatives and percentage rates of return. [3]

	States of Nature (Market Conditions)		
	Low	Medium	High
Regular shares	7%	10%	15%
Risky shares	-10%	12%	25%
Property	-12%	18%	30%

Over the past 300 days, 150 days have been medium market conditions and 60 days have had high market increases. On the basis of these data, state the optimum investment strategy for the investment.

OR

- a) A company is currently involved in negotiations with its union on the upcoming wage contract. Positive signs in table represent wage increase while negative sign represents wage reduction. What are the optimal strategies for the company as well as the union? What is the game value? [7]

Conditional costs to the company (Rs. in lakhs)

		Union Strategies			
		U ₁	U ₂	U ₃	U ₄
Company Strategies	C ₁	0.25	0.27	0.35	-0.02
	C ₂	0.20	0.06	0.08	0.08
	C ₃	0.14	0.12	0.05	0.03
	C ₄	0.30	0.14	0.19	0.00

- b) A manufacturer, manufactures a product, of which the principal ingredient is a chemical X. At the moment, the manufacturer spends Rs. 1,000 per year on supply of X, but there is a possibility that the price may soon increase to four times its present figure because of a worldwide shortage of the chemical. There is another chemical Y, which the manufacturer could use in conjunction with a third chemical Z, in order to give the same effect as chemical X. Chemicals Y and Z would together cost the manufacturer Rs. 3,000 per year, but their prices are unlikely to rise. What action should the manufacturer take using minimax criteria for decision-making? [3]

